FFT Design Report

1. Introduction

Fast Fourier Transform (FFT) is an efficient algorithm to compute the Discrete Fourier Transform (DFT) and its inverse. The FFT algorithm reduces the computational complexity of DFT from O(N^2) to O(N log N), making it highly efficient for signal processing applications.

2. RTL Design

fft_8pt.v & fft_8pt_Tb.v

Tools Used: Vivado

Description:

The RTL design for the FFT module includes the following sub-modules:

- Butterfly Processing Unit

- Twiddle Factor Multiplication

- Bit-Reversal Logic

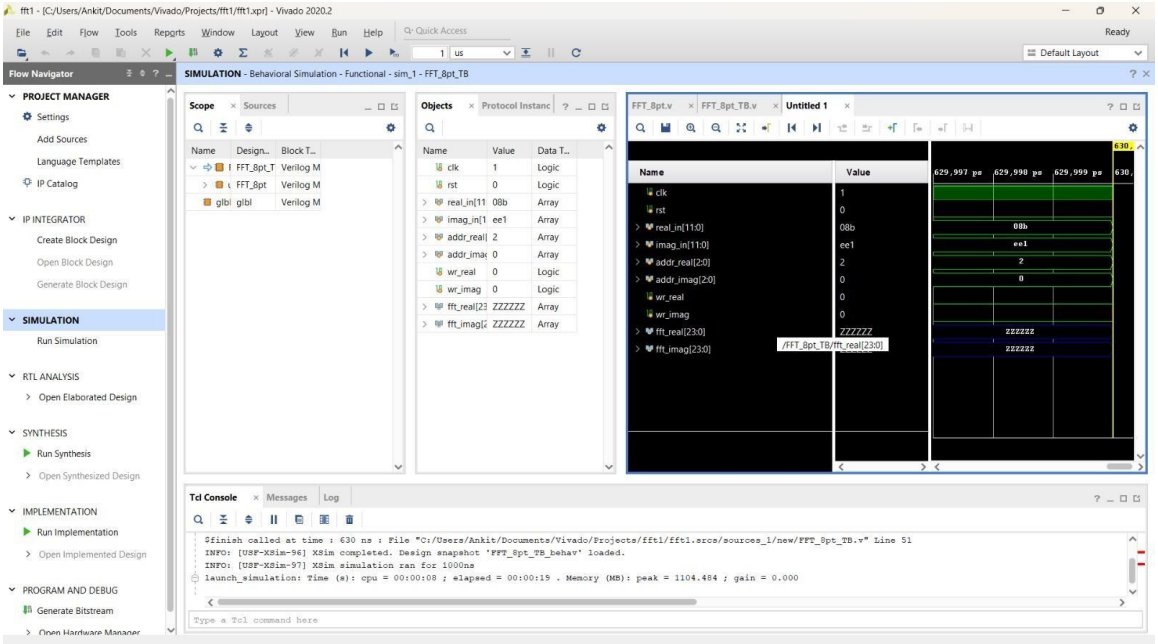
- Control FSM for stage and data flow management

Constructs Observed:

- Arithmetic operations (adders, multipliers) inferred from fixed-point logic

- FSM inferred from case statements and if-else structures

- Registers and logic inferred from always_ff and always_comb

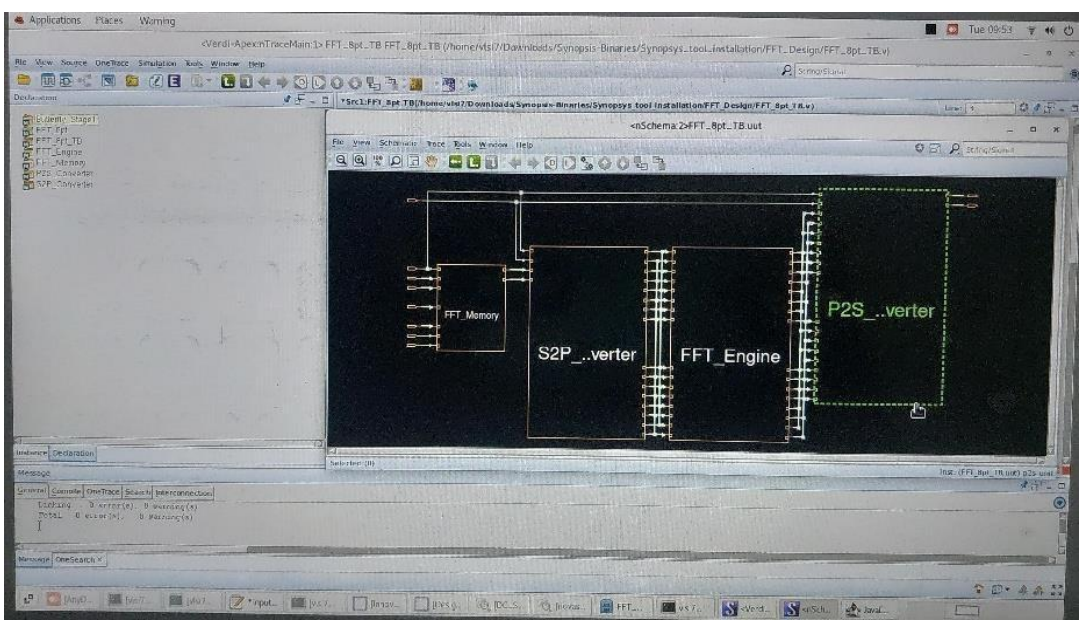


3. Pre-Synthesis Simulation

Tool Used: Synopsys VCS, Verdi , nclaunch

Description:

The pre-synthesis simulation validates the functional correctness of the FFT RTL design, ensuring all logic paths and data flows are accurately represented before synthesis.

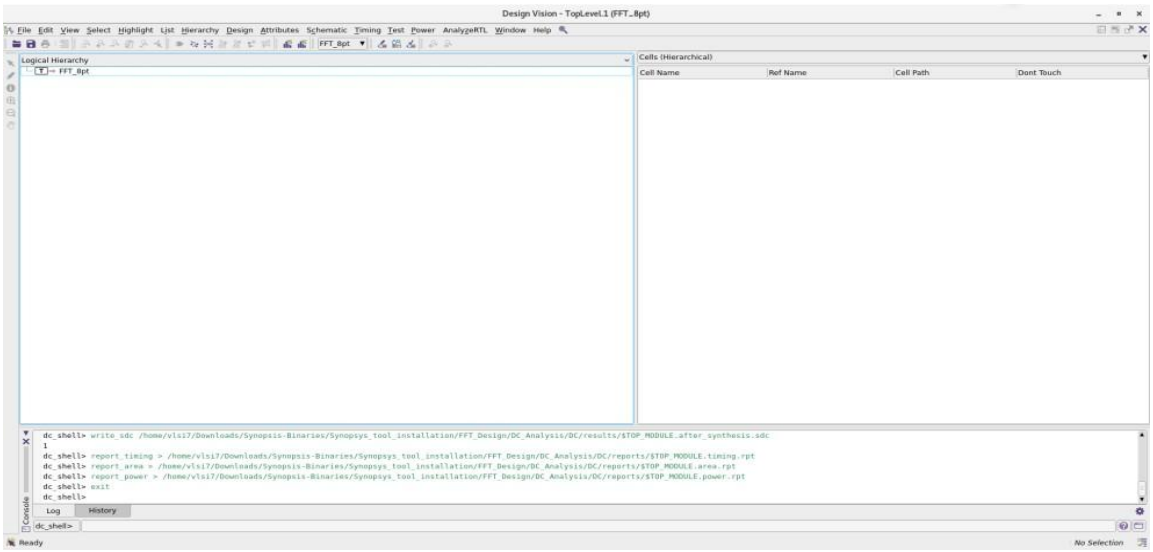


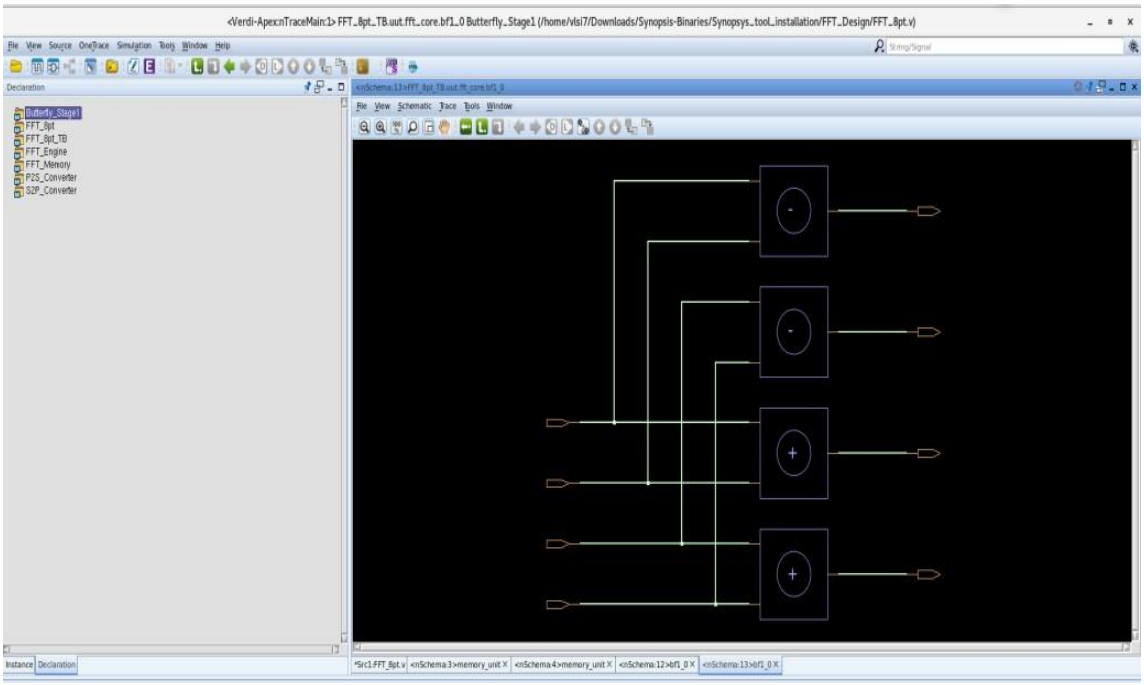
4. Synthesis

Tool Used: Synopsys Design Compiler

Description:

The RTL is synthesized into a gate-level netlist, applying timing constraints and optimizations. Key parameters include clock constraints, area optimization, and critical path analysis.



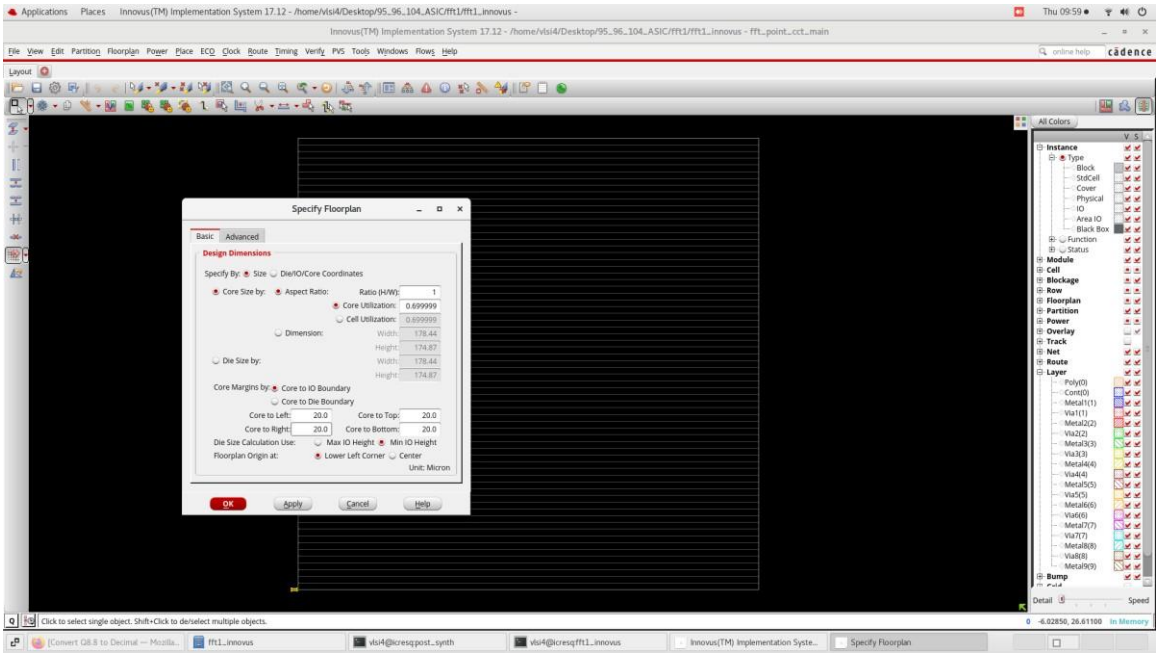


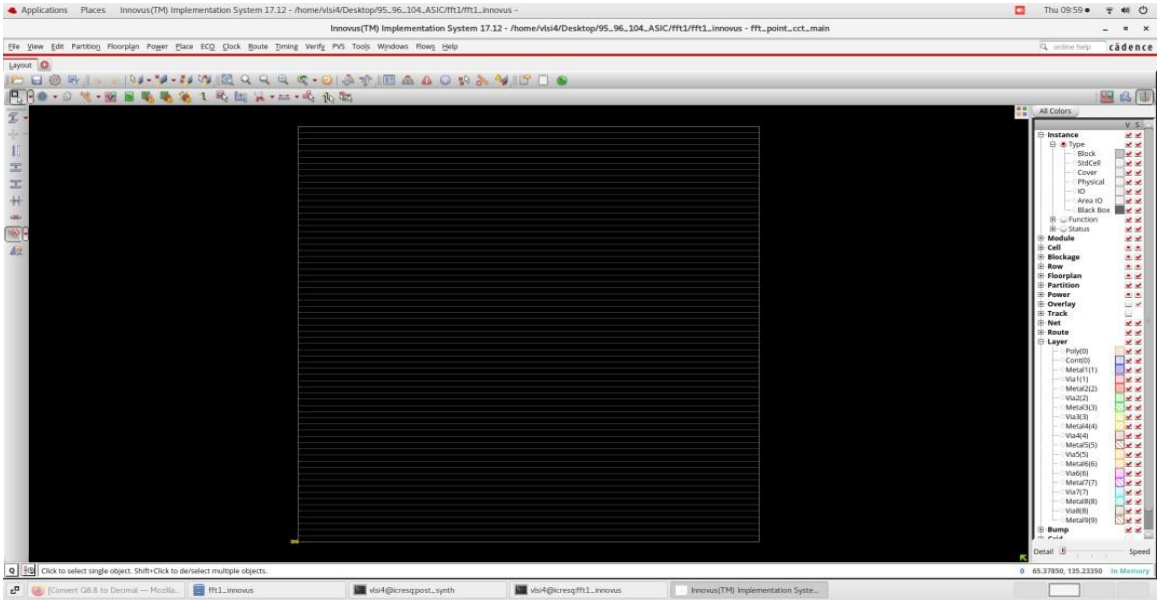
5. Floor planning

Tool Used: Cadence Innovus

Description:

The initial placement of macros, I/O pads, and standard cells is defined, optimizing the core area utilization and routing paths.



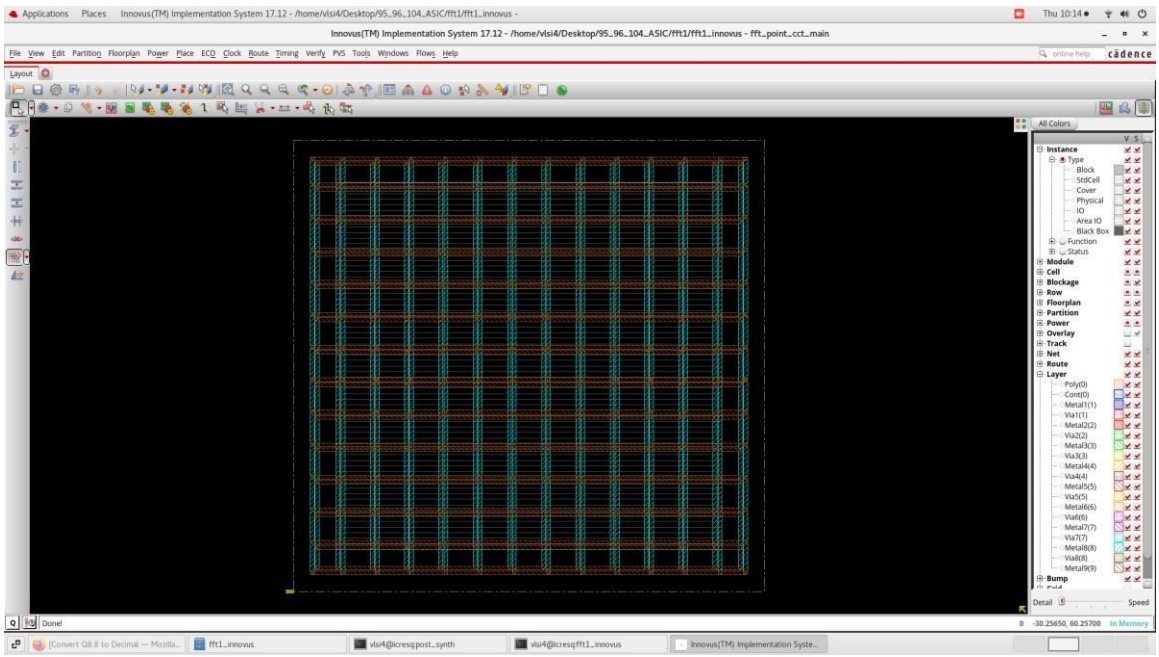


6. Power Planning

Tool Used: Innovus

Description:

Power rings and stripes are defined to ensure even power distribution across the design, preventing IR drops and electromigration.



7. Placement

Tool Used: Innovus

Description: Placement optimizes the locations of standard cells, macros, and I/O pads for minimal wire length and optimal timing.

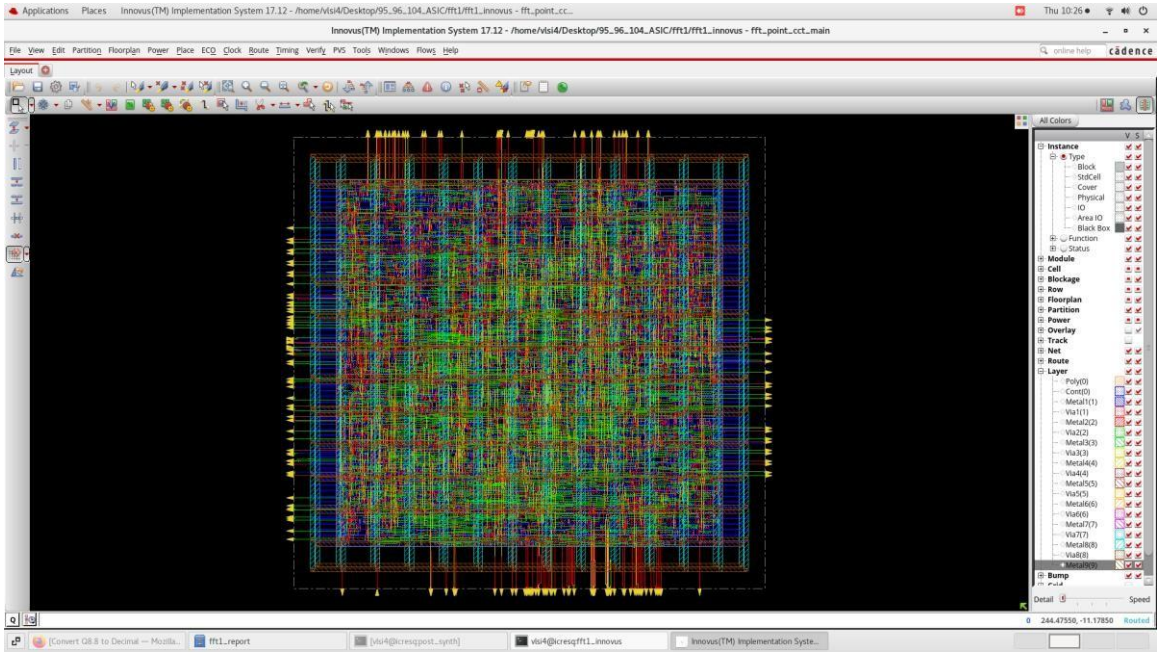
Steps Involved:

1. **Global Placement:** Initial cell layout.
2. **Detailed Placement:** Overlap removal and optimization.
3. **Legalization:** Ensures cells follow design rules.

Commands:

- place_opt: Optimize placement.
- legalize: Legalize cell positions.

- check_placement: Verify placement.



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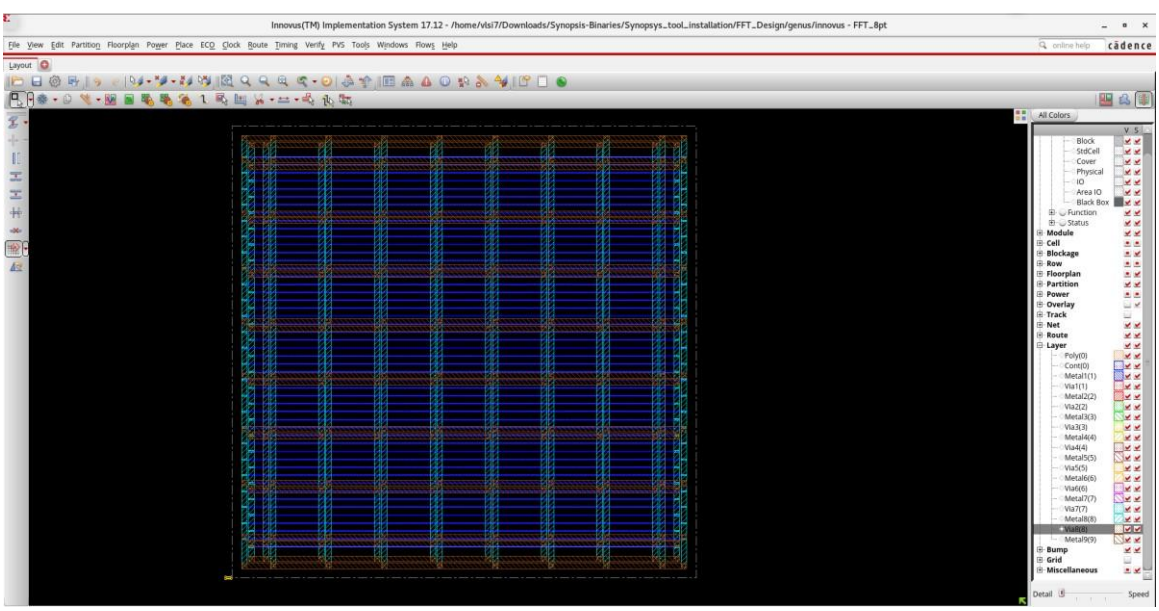
8. Routing

Tool Used: Innovus

Description:

Routing of the design is completed with special attention to clock tree synthesis and signal integrity.



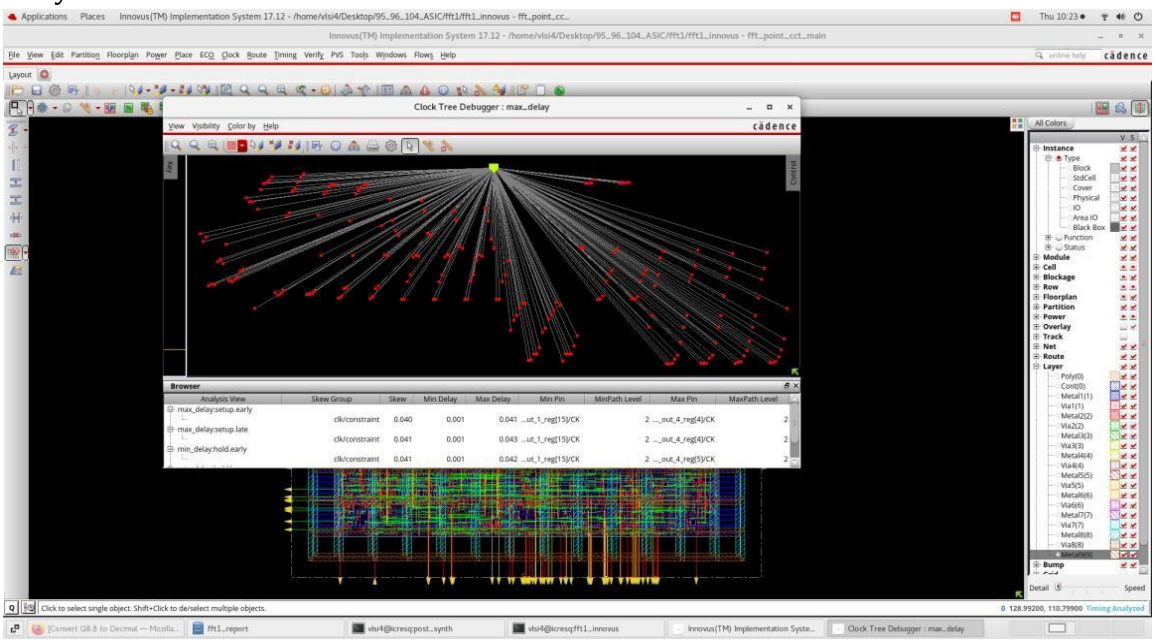
9. Clock Tree Synthesis (CTC)

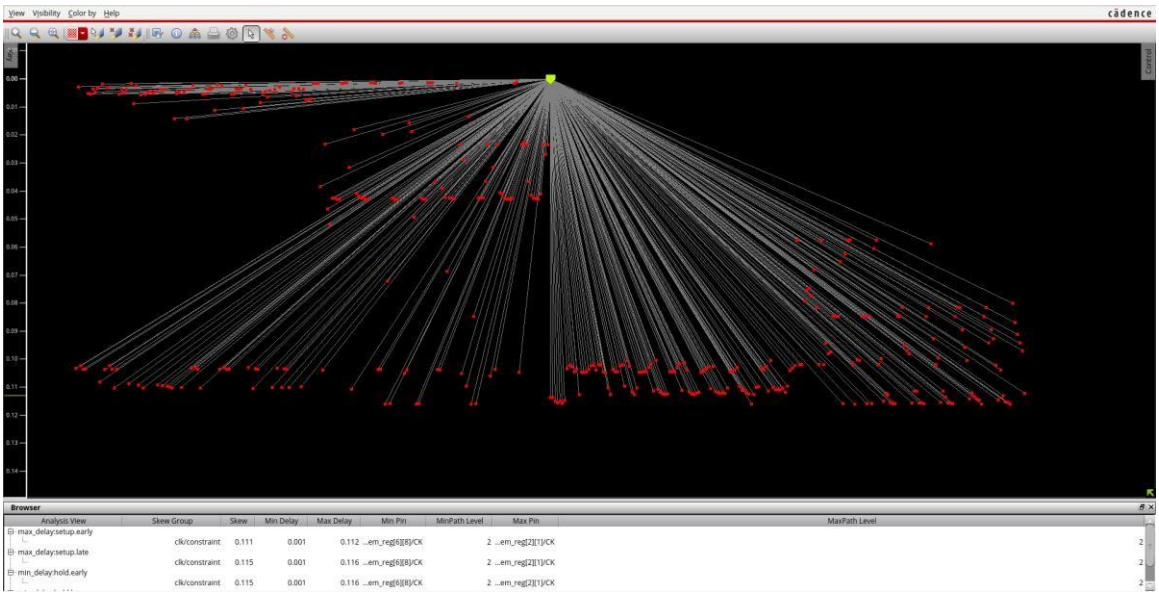
Tool Used: Innovus

Description: Clock Tree Synthesis (CTS) is performed to distribute the clock signal evenly across all sequential elements, minimizing skew and ensuring synchronization.

Observed Structure: The clock tree consists of buffer stages and clock gating cells, optimized to reduce latency and maintain timing integrity.

Analysis: During CTS, constraints such as clock uncertainty and latency are adjusted to optimize timing closure. Varying these parameters impacts the clock skew and timing path delay.



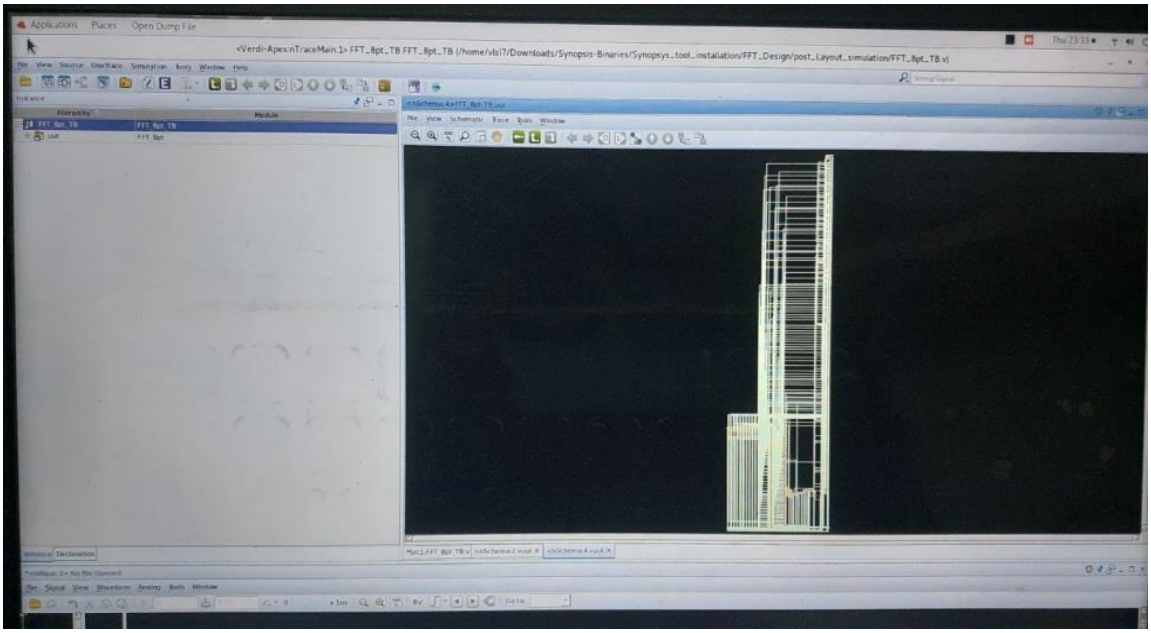
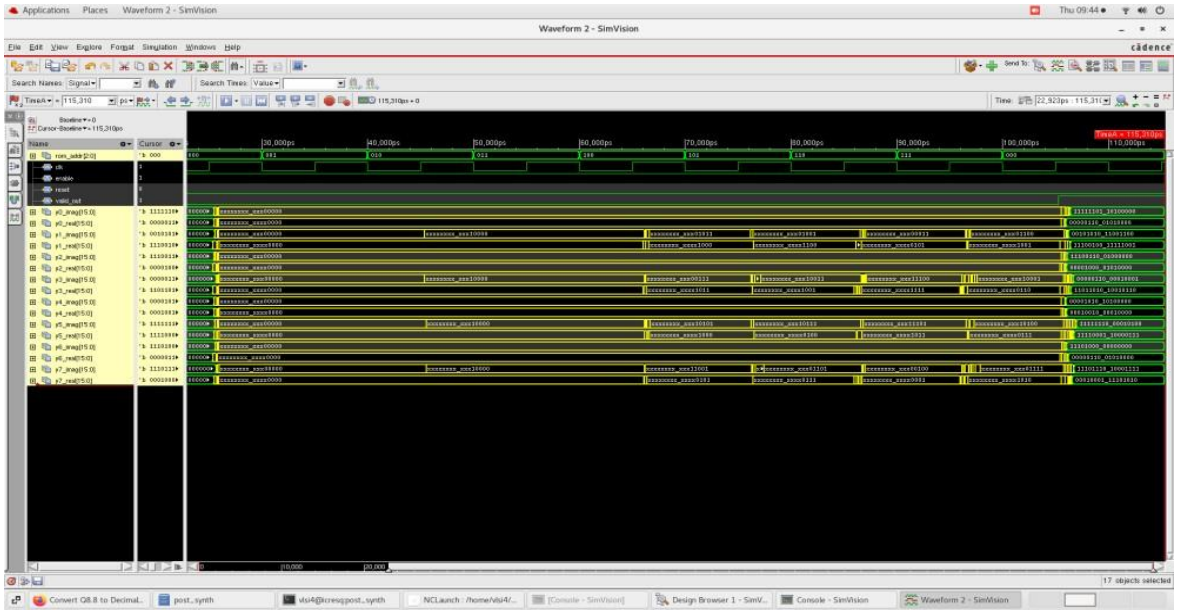


10. Post-Synthesis Simulation

Tool Used: NCLaunch

Description:

Simulation is performed on the synthesized netlist to verify timing correctness and functional integrity.



dc-shell schematic